

CHALLAN SETTLEMENT SYSTEM

Project Understanding & System Documentation

1 Introduction

The Challan Settlement System is an Android-based mobile application that helps users settle their vehicle challans through a managed court settlement service.

Instead of paying the full challan amount directly, users pay a reduced settlement amount through the app. The remaining process is handled by our physical party team who visits the court and clears the challan on behalf of the user.

The settlement process takes up to 15 days from the date of payment.

2 Problem Statement

Many vehicle owners face the following issues:

- Multiple challans on their vehicle
- High total challan amount
- No time to visit court
- Complex legal procedure
- Long queues and paperwork

Our system solves these problems by:

- Providing a simple mobile interface

- Showing total challan amount
 - Offering a settlement amount
 - Handling court process through physical party
 - Updating status within 15 days
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3 Business Model

Example Scenario:

Total Challan Amount = ₹6000

Settlement Offer = ₹5000

User pays ₹5000 through PhonePe.

Our physical party handles the full ₹6000 challan clearance in court.

User does not need to visit court.

Settlement Time: Up to 15 working days.

4 System Architecture

The system consists of three main parts:

◆ 1. Frontend (React Native Android App)

Built using React Native.

Handles:

- User input

- Vehicle number entry
 - Display RC details
 - Display challan details
 - Show settlement amount
 - Payment integration
 - Status tracking
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◆ 2. Backend (Django + REST API)

Built using Django.

Responsible for:

- Fetching vehicle details
 - Fetching challan data
 - Calculating settlement amount
 - Storing payment records
 - Managing 15-day workflow
 - Admin panel management
 - Assigning physical party
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◆ 3. Admin Panel (Django Admin or Custom Panel)

Used by internal team (backend team) .

Admin can:

- View users
 - View vehicles
 - View challans
 - View settlement payments
 - Assign physical party
 - Upload court receipt
 - Update case status
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5 Complete User Workflow

Below is the full user journey:

Step 1: User Enters Vehicle Number

User opens app and enters vehicle plate number.

Example:

RJ14AB1234

Step 2: System Fetches Data

Backend fetches:

- RC Details
- All pending challans
- Total challan amount

Example:

Challan No	Amount
CH001	₹1000
CH002	₹2000
CH003	₹3000

Total = ₹6000

Step 3: Settlement Offer Display

Instead of asking user to pay ₹6000, app shows:

"Settle all challans for ₹5000"

User sees:

- Total Challan Amount
 - Settlement Amount
 - 15 Days Processing Notice
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Step 4: Payment

User pays ₹5000 using PhonePe.

After successful payment:

Status = Settlement Initiated

Processing Time = Up to 15 Days

Step 5: 15-Day Settlement Workflow

After payment:

Day 1-2:

- Case registered in admin panel

Day 3-10:

- Physical party visits court
- Pays full challan amount
- Collects official receipt

Day 11-15:

- Admin verifies payment
- Uploads receipt
- Marks challan as cleared

User receives status update:
"Your Challan Has Been Cleared"

6 Settlement Status Flow

Stage	Status
Before payment	Awaiting Payment
After user pays	Settlement Initiated
Assigned to physical party	In Court Processing
Court payment completed	Under Verification
After verification	Cleared

7 Admin Workflow

Admin Dashboard includes:

1. Dashboard Overview

- Total Users
 - Total Vehicles
 - Total Cases
 - Cases In Progress
 - Cleared Cases
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2. Case Management

Admin can:

- View vehicle details
 - See total challan
 - See settlement amount paid
 - Assign case to physical party
 - Update status
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3. Physical Party Assignment

Admin assigns:

- Physical party name
- Court location

- Due date (within 15 days)
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4. Receipt Upload

After court payment:

Admin uploads:

- Court receipt image
 - Payment confirmation
 - Clearance date
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8 Database Structure

♦ User Table

- id
- name
- mobile
- email

♦ Vehicle Table

- id
- user_id
- vehicle_number
- rc_details

◆ **Challan Table**

- id
- vehicle_id
- challan_number
- amount
- status

◆ **Settlement Table**

- id
- vehicle_id
- total_challan_amount
- settlement_amount
- payment_status
- court_status
- assigned_physical_party
- payment_date
- expected_clearance_date
- receipt_file

9 Frontend Screens (React Native)

1. Splash Screen

2. Login / Register
 3. Vehicle Entry Screen
 4. Challan Details Screen
 5. Settlement Offer Screen
 6. Payment Screen (PhonePe)
 7. Processing Screen (15 Days Notice)
 8. Settlement Status Screen
 9. Payment History Screen
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Technical Stack

Layer	Technology
Mobile App	React Native
Backend	Django REST Framework
Database	PostgreSQL
Payment Gateway	PhonePe
Admin Panel	Django Admin

Important System Rules

1. User cannot partially pay.
2. User always pays fixed settlement amount.

3. Full challan clearance happens offline in court.
 4. Maximum processing time = 15 days.
 5. Admin must update status manually.
 6. Receipt must be uploaded for confirmation.
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Security Considerations

- Secure payment gateway integration
 - Transaction ID storage
 - Admin authentication
 - Data encryption
 - API authentication using tokens
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Future Enhancements

- SMS updates
 - Push notifications
 - Automatic RTO integration
 - Live status tracking
 - Multi-state support
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Conclusion

The Challan Settlement System is a complete digital-to-physical service model.

It combines:

- Mobile technology (React Native)
- Backend APIs (Django)
- Online payment (PhonePe)
- Offline court settlement (Physical Party)
- Admin management panel
- 15-day structured processing system

This solution makes challan settlement easy, fast, and hassle-free for vehicle owners.